

Semahattin Can Doruk

Software Developer (Flutter • C# • Embedded Systems)

Phone: TR +90 531 255 4069 [Email: semahattincandoruk@gmail.com](mailto:semahattincandoruk@gmail.com) [Website: https://candorukk.github.io](https://candorukk.github.io)
[GitHub: https://github.com/CanDorukk](https://github.com/CanDorukk) [LinkedIn: https://www.linkedin.com/in/candorukk/](https://www.linkedin.com/in/candorukk/)



PROFESSIONAL SUMMARY

Software Developer specialized in designing and building end-to-end architectures across mobile, desktop, and embedded systems. Highly experienced in C#, Flutter, Dart, and Python for cross-platform development, IoT infrastructure, hardware-software integration, and Linux customization. Skilled in refactoring incomplete projects for stability, implementing bidirectional serial protocols, and building control interfaces for custom hardware.

TECHNICAL SKILLS

- **Programming Languages:** Dart (Intermediate-Advanced), C# (Intermediate-Advanced), Python (Intermediate-Advanced)
- **Mobile & Desktop Development:** Flutter (Advanced), Flutter Desktop (Linux/Windows), C# WPF, UI/UX Design (Intermediate)
- **Embedded Systems & Hardware:** ESP32, Raspberry Pi, Jetson Orin Nano, ROS, CMake, Serial Communication, Bidirectional Communication Protocols
- **Backend & Database:** Supabase (RLS, Auth), PostgreSQL, SQLite (Intermediate), REST API Integration
- **Operating System & Integration:** Linux Boot Customizations, Embedded Lock/Security Controls, Kiosk Mode Configurations
- **Data Science & Artificial Intelligence:** Data Science (Intermediate), Machine Learning (Intermediate)
- **Design & Office Tools:** Canva (Intermediate), Photoshop (Intermediate-Advanced), MS Excel (Advanced), MS Word (Advanced)
- **Languages & Skills:** English (C1), Teamwork, Problem Solving, Fast Adaptation, Analytical Thinking, Responsibility & Initiative

JOB EXPERIENCE

Software Developer — SANCAKTAR Mekatronik Sistemler | 07.2025 – 03.2026

- Solely took over an incomplete Bioreactor Monitoring Software initially started with C# WPF, completed missing and dysfunctional modules, and redesigned system lock and security controls from scratch to achieve field-ready stability.
- Implemented MVVM architecture and multithreading optimization to enable real-time sensor data monitoring, error logging, and high UI responsiveness.
- Established a bidirectional serial communication (UART/Serial) infrastructure and database integrations to ensure data integrity between the embedded control board and desktop application.
- Migrated the project to Flutter Desktop architecture to achieve cross-platform support across Windows & Linux, subsequently leading parallel development workflows within a 2-person team.
- Performed Linux boot customizations to display company branding on system startup, and integrated hardware/software locking mechanisms to restrict external device connections except for authorized flash drives for system security and hardware compliance.

Software Developer (Project-Based) — UGF Makine Sanayi ve Ticaret A.Ş. | 07.2025 – 01.2026

- Smart Sprinkler System Project: Architected, designed, and developed the end-to-end infrastructure independently across hardware (ESP32), mobile application (Flutter), and backend (Supabase).
- Configured an Access Point (Wi-Fi AP) mode on the ESP32 for initial setup, allowing users to connect directly via the mobile app to configure local network settings.
- Built end-to-end integration by implementing a Supabase and PostgreSQL database schema, user authentication, real-time data streaming, and backend logic.
- Weeding Robot: Managed the software integration and designed the control interface layouts for the agricultural weeding robot.

Software Developer Intern — Nefel Hassas Tarım | 06.2025 – 08.2025

- Participated in software development and integration workflows for Smart Solar Panel Cleaning Robots and Autonomous Agricultural Robots.
- Handled setup, configuration, and hardware-software integration of ROS (Robot Operating System) environments on Jetson Orin Nano and Raspberry Pi platforms.
- Utilized and implemented CMake build systems to manage project compilation and configuration workflows.
- Integrated serial communication protocols for robot motion control and designed sensor-based headland turn navigation algorithms for field operation.
- Authored technical documentation covering system installation and configuration steps to streamline field deployment procedures.

TEDx Alparslan University | Team Member

- Provided operational support during event organization processes and managed inter-community communication.

Kodland | Python Instructor (Part-Time) | 04.2024 – 07.2024

- Delivered training on Python, algorithmic logic, and fundamental software concepts to primary and high school students.

PERSONAL PROJECTS

- **Habitly Daily:** A mobile application for habit tracking, reminders, and focus sessions. (Flutter, Riverpod, Hive, AdMob, In-App Purchase)
- **CrestBalance:** An Erasmus-focused financial management app enabling multi-currency budget and grant tracking. (Flutter, Supabase, PostgreSQL, Riverpod)
- **Voice2 Note AI:** A smart note-taking application converting audio recordings into text and generating automated summaries. (Flutter, FastAPI, SQLite, faster-whisper)
- **AgroMora:** A mobile e-commerce application integrated with role-based access control (RBAC). (Flutter, Supabase, PostgreSQL)
- **QuizArea & ExploreLab & DevWeave:** Educational platforms covering English learning, content-automated interactive learning, and software documentation/training.

EDUCATION

- Muş Alparslan University | B.Sc. in Information Systems and Technologies — GPA: 3.10/4.00 (2022 – 2026)
- University of Beira Interior (Portugal) | Erasmus+ Exchange Program (Spring 2026)
- Yahya Günsür Vocational and Technical Anatolian High School | Web Design and Programming (2016 – 2020)